Digital Signal Processing

Digital Signal Processing

Chapter 2 – The Z-Transform and its application to the analysis of LTI systems

- **Z- Transform.**
- **Region of Convergence of Z-Transform.**
- **Inverse Z-Transform.**
- **solution of difference equations using one sided Z-transform.**

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2 Z-Transform

- The Direct z-Transform
  - The ZT is a transformation that maps DT signal x[n] into a function of the complex variable z, defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

  - Notation:

$$X(z) \equiv Z\{x(n)\}$$

$$x(n) \overset{z}{\leftrightarrow} X(z)$$

  - z is a complex variable that can be represented as $z = r\, e^{j\omega}$.
  - Substituting $z = e^{j\omega}$ will reduce the z-transform to discrete-time Fourier transform(DTFT) to be covered in other chapters.

$$X\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

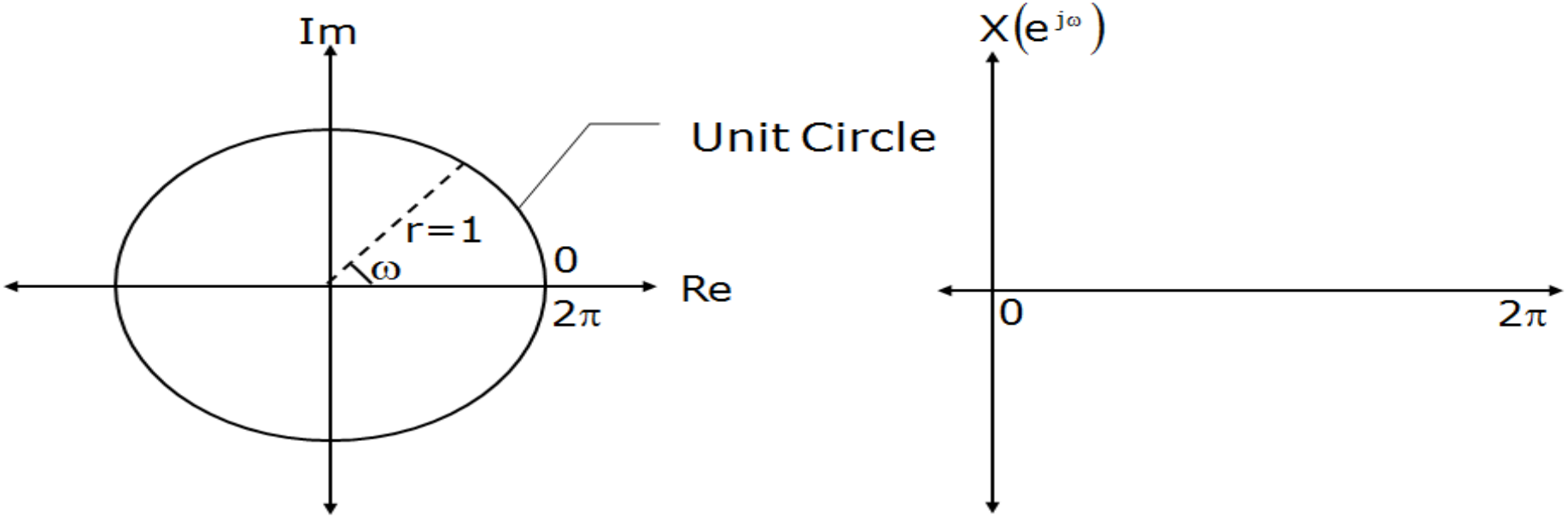
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2 Z-Transform

2.1.the direct Z-Transform

Cont'd

- The z-transform is a function of the complex z variable.
- Convenient to describe on the complex z-plane.
- If we plot **$z = e^{j\omega}$ for ω=0 to 2π** we get the unit circle.





Z-Transform

2.2. Region of Convergence

- The z-transform does not converge for all sequences or for all values of z —the infinite sum may not always be finite.
- The **region of convergence (ROC)** of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.
- The z-Transform is, therefore, uniquely characterized by:
  1. expression for X(z)
  2. ROC of X(z)



2.2. Region of Convergence

Cont'd

- The z-transform therefore exists (or converges) if

$$\sum_{-\infty}^{\infty} |x[n]||z|^{-n} < \infty$$

  - If some value of z, say, $\mathbf{z = z_1}$, is in the **ROC**, then **all values of z** on the circle defined by $|z| = |z_1|$ will also **be in the ROC**.
  - So, the ROC will consist of a ring in the z-plane centered about the origin. Its **outer boundary** will be a circle (*or the ROC may extend outward to infinity*), and its **inner boundary** will be a circle (*or it may extend inward to include the origin*).

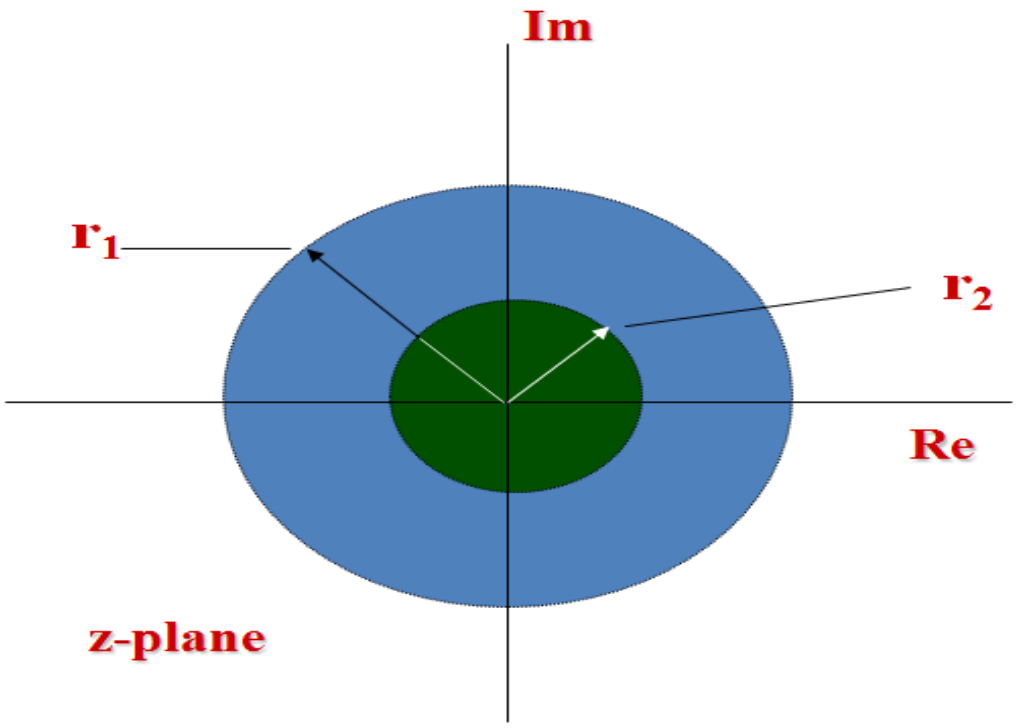
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2 Z-Transform

2.2. Region of Convergence

Cont'd

- The z-transform and all its derivatives must be continuous functions of z within the ROC.
- X(z) can be represented in a Laurent (one type of power) series and a rational function of polynomial in z.
- The ROC of a causal signal is the exterior of a circle of some radius r2 while the ROC of an anti-causal is the interior of a circle of some radius r1.



2

Properties of the ROC

- The ROC is *always bounded* by a circle since the convergence condition is on the magnitude |z|;
- The ROC for right-sided sequences ($n<n_0$, $x(n)=0$) is *always outside of a circle of radius* R_{x-}. (if $n_0>=0$, x(n) is a causal seq.)
- The ROC for left-sided sequences ($n>n_0$, $x(n)=0$) is *always inside of a circle of radius* R_{x+}. (if $n_0<=0$, x(n) is a anticausal sequence)
- The ROC for two-sided sequences is *always an open ring* $R_{x-}<|z|<R_{x+}$ if it exists.
- The ROC for *finite-duration* sequences (n<n1 and n>n2, x(n)=0) is the entire z-plane. If $n_1<0$, then z=infinity is not in the ROC. If $n_2>0$, then z=0 is not in the ROC;
- The ROC cannot include a pole since X(z) converges uniformly in there;
- There is *at least one pole* on the boundary of a ROC of a rational X(z);
- The ROC is one contiguous region, the ROC does not come in pieces.

4

The complex variable z can be written in polar form as

$$z = re^{j\theta}$$

The X(z) can be expressed as

$$X(z)\Big|_{z=re^{j\theta}} = \sum_{n=-\infty}^{\infty} x(n) r^{-n} e^{-j\theta n}$$

$$\begin{aligned} |X(z)| &= \left| \sum_{n=-\infty}^{\infty} x(n) r^{-n} e^{-j\theta n} \right| \le \sum_{n=-\infty}^{\infty} \left| x(n) r^{-n} \right| < \infty \\ &\le \sum_{n=-\infty}^{-1} \left| x(n) r^{-n} \right| + \sum_{n=0}^{\infty} \left| \frac{x(n)}{r^n} \right| = \sum_{n=1}^{\infty} \left| x(-n) r^{n} \right| + \sum_{n=0}^{\infty} \left| \frac{x(n)}{r^n} \right| \end{aligned}$$

4

In the first sum, there must be exist values of r small enough so that $x(-n)r^n$ is absolutely summable.

In the second sum, there must exist values of r large enough so that $x(n)/r^n$ is absolutely summable.

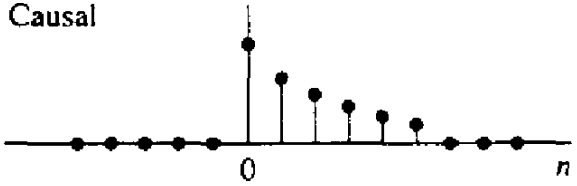
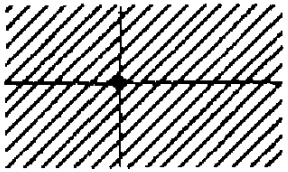
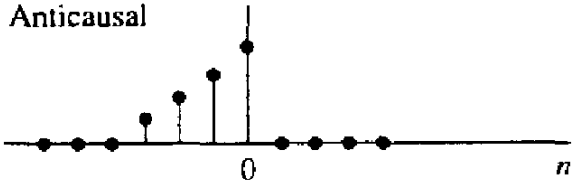
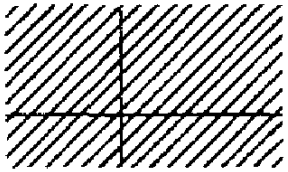
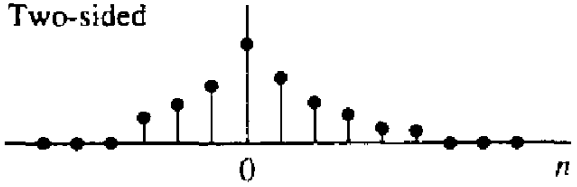
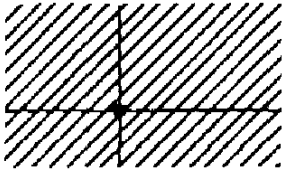
Since the convergence of X(z) requires that both sums, the ROC of X(z) is specified the common region where the both sum are finite.

If there is no common region , then X(z) does not exist.

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2 Z-Transform

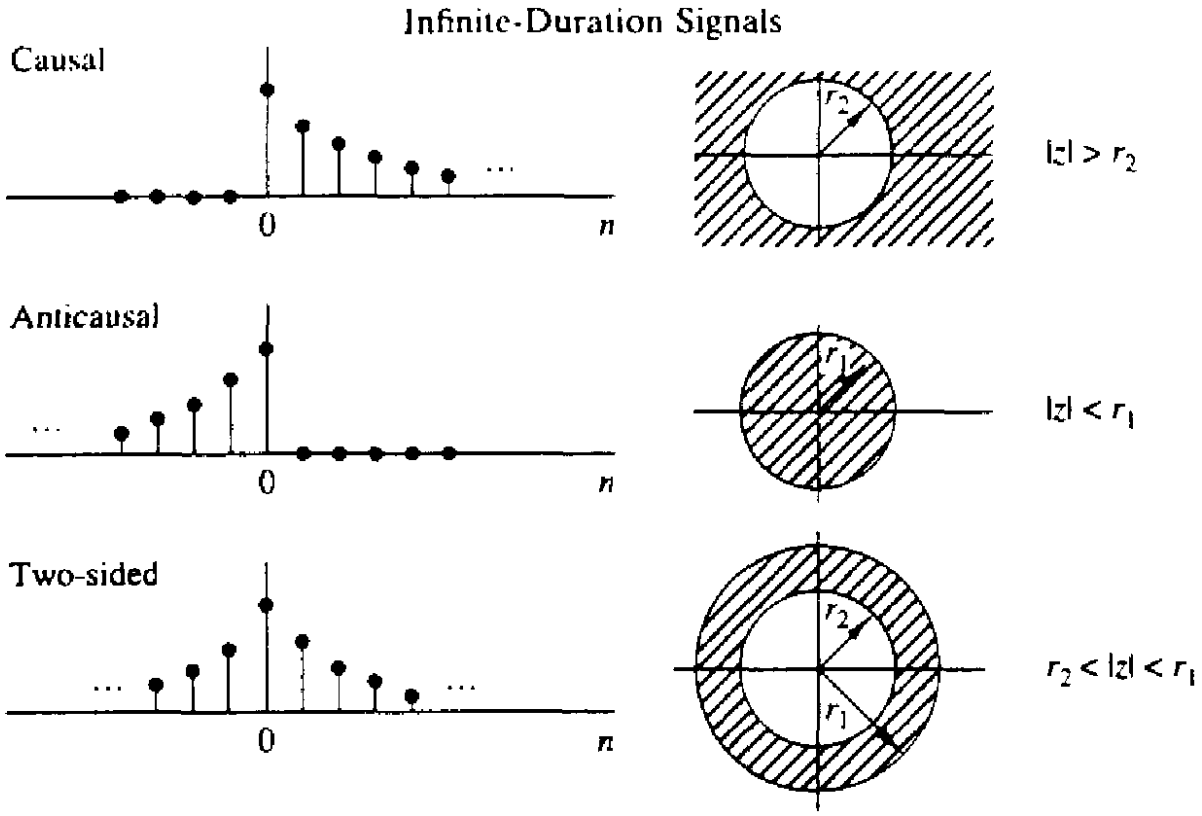
2.2. Region of Convergence

Signal	ROC
Finite-Duration Signals	
Causal 	 Entire z-plane except $z = 0$
Anticausal 	 Entire z-plane except $z = \infty$
Two-sided 	 Entire z-plane except $z = 0$ and $z = \infty$

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2 Z-Transform

2.2. Region of Convergence



Digital Signal Processing

2 Z-Transform

2.2. Region of Convergence

Common Z-Transform Pairs:

Table 3.1 Some common z-transform pairs

	Sequence $x[n]$	z-Transform $X(z)$	ROC
1.	$\delta[n]$	1	All z
2.	$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
3.	$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
4.	$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
5.	$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
6.	$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
7.	$(\cos \omega_0 n)u[n]$	$\frac{1-(\cos\omega_0)z^{-1}}{1-2(\cos\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
8.	$(\sin \omega_0 n)u[n]$	$\frac{(\sin\omega_0)z^{-1}}{1-2(\cos\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
9.	$(r^n \cos \omega_0 n)u[n]$	$\frac{1-(r\cos\omega_0)z^{-1}}{1-2(r\cos\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
10.	$(r^n \sin \omega_0 n)u[n]$	$\frac{(\sin\omega_0)z^{-1}}{1-2(r\cos\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$

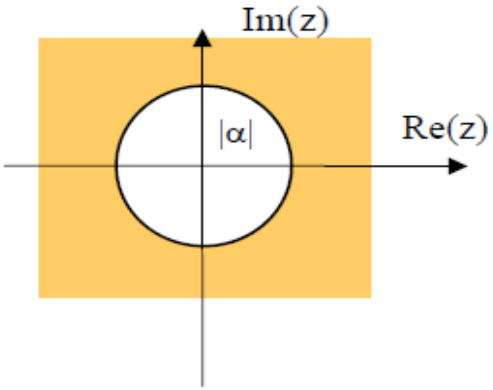
2

Example.1:

$$x(n)=\alpha^n u(n)=\begin{cases}\alpha^n & n\geq 0\\ 0 & n<0\end{cases}$$

Find z-transform of x(n)

$$X(z)=\sum_{n=0}^{\infty}\alpha^n z^{-n}=\sum_{n=0}^{\infty}(\alpha z^{-1})^n$$



if $|\alpha z^{-1}|<1$ the power series is

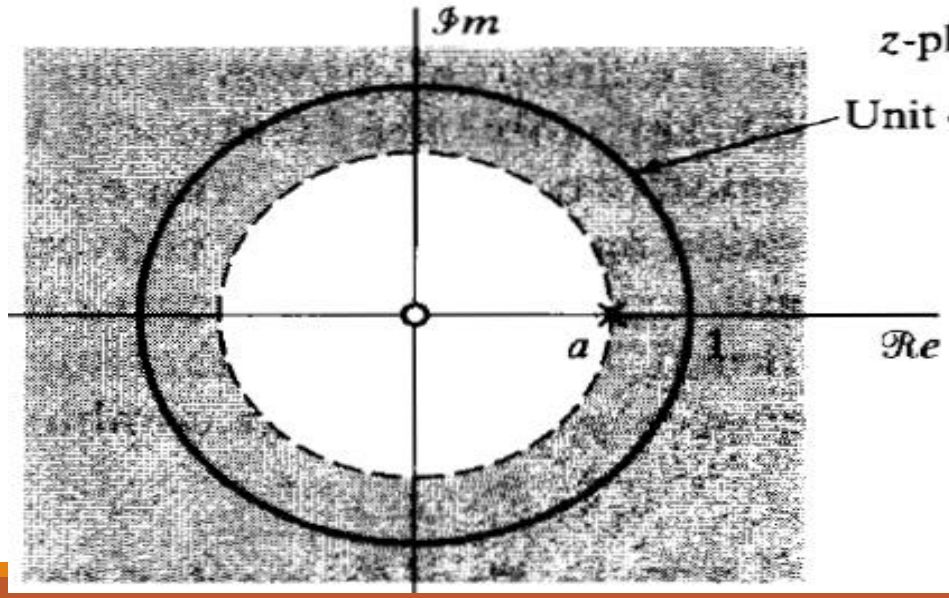
$$X(z)=\frac{1}{1-\alpha z^{-1}}$$

ROC: $|z|>|\alpha|$

2

Inside the region of convergence, infinite series converges to

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \qquad |z| > |a|.$$



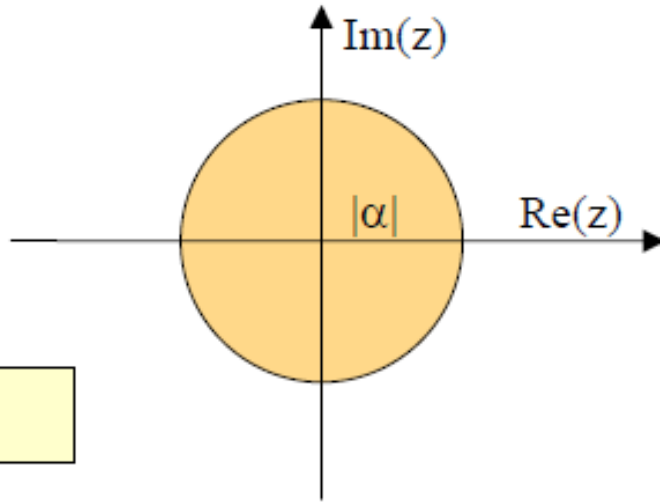
Pole-zero plot and region of convergence

2

Example.1: Find z-transform of x(n)

$$x(n) = -\alpha^n u(-n-1) = \begin{cases} 0 & n \geq 0 \\ -\alpha^n & n \leq -1 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n) z^{-n} = -\sum_{m=1}^{\infty} (\alpha^{-1} z)^m$$

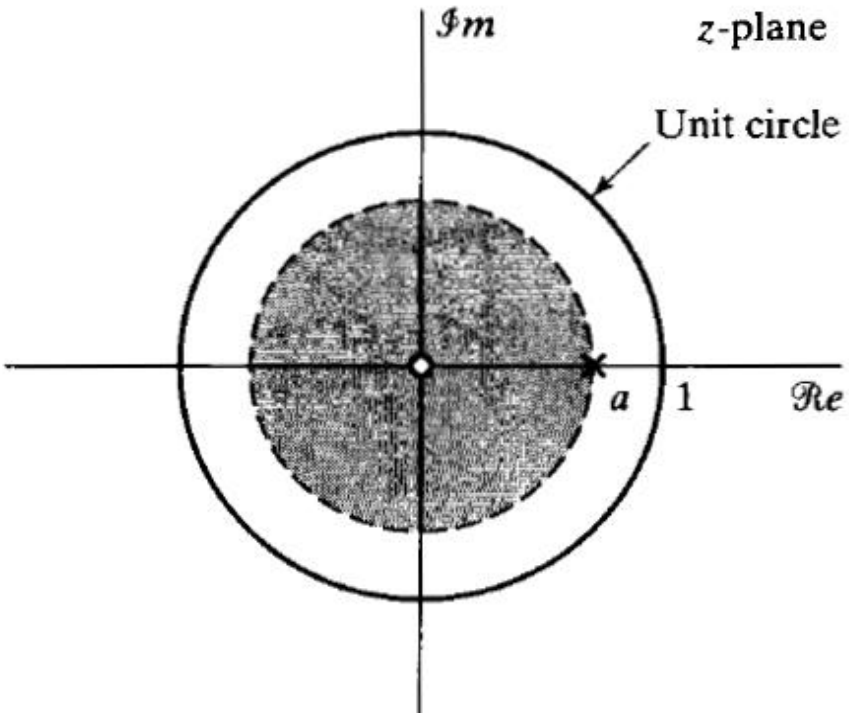


if $|\alpha z^{-1}| < 1$ the power series is

$$X(z) = -\frac{\alpha^{-1} z}{1 - \alpha^{-1} z} = \frac{1}{1 - \alpha z^{-1}}$$

ROC: $|z| < |\alpha|$

2

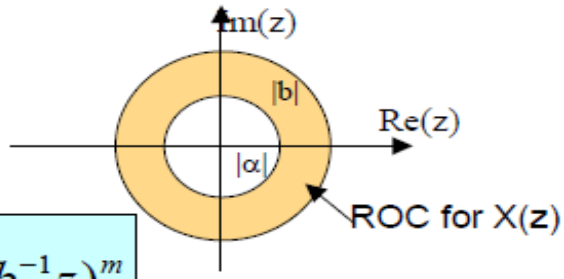


Pole-zero plot & region of convergence(ROC)

2

Example.1: Find z-transform of x(n)

$$x(n) = \alpha^n u(n) + b^n u(-n-1)$$



$$X(z) = \sum_{n=0}^{\infty} (\alpha)^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n + \sum_{m=1}^{\infty} (b^{-1} z)^m$$

From the first power series if $|\alpha z^{-1}| < 1$ or $|z| > |\alpha|$ and from the second power series if $|b^{-1} z| < 1$ or $|z| < |b|$, then

We have two cases:

$|b| < |\alpha|$ There is no common region, X(z) does not exist.

$|b| > |\alpha|$ There is common region, which is , Then we obtain

$$X(z) = \frac{1}{1 - \alpha z^{-1}} - \frac{1}{1 - b z^{-1}} = \frac{b - \alpha}{(1 - \alpha z^{-1})(1 - b z^{-1})}$$

ROC $|\alpha| < |z| < |b|$

2

- The ROC cannot include any poles.
- The ROC is a connected (that is, a single contiguous) region.
- For finite duration sequences the ROC is the entire z-plane, with the possible exception of $z = 0$ or $z = \infty$.
- For infinite duration sequences the ROC can have one of the following shapes:

Type of sequence		ROC
Right-sided ($x[n] = 0, n < n_0$)	$\Rightarrow$	ROC: $\lvert z\rvert > r$
Left-sided ($x[n] = 0, n > n_0$)	$\Rightarrow$	ROC: $\lvert z\rvert < r$
Two-sided	$\Rightarrow$	ROC: $a < \lvert z\rvert < b$

- To uniquely specify a sequence $x[n]$ we need *both* $X(z)$ and its ROC.
- The function $X(z)$ is legitimate only for z within its ROC. $X(z)$ is not defined when z is outside the ROC, even if the formula for $X(z)$ yields meaningful results for these values.

2 Z-Transform

2.2. Examples

I. Determine the z-transforms of the following finite-duration signals.

a) $x_1(n) = \{1,2,5,7,0,1\}$

$X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$, ROC: entire z-plane except z = 0.

b) $x_2(n) = \{1,2,\underset{\uparrow}{5},7,0,1\}$

$X_2(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$, ROC: entire z-plane except z = 0 & z = ∞.

c) $x_3(n) = \delta(n)$

$X_3(z) = 1$, ROC: entire z-plane.

d) $x_4(n) = \delta(n-k)$, $k > 0$.

$X_4(z) = z^{-k}$, $k > 0$, ROC: entire z-plane except z = 0.

e) $x_5(n) = \delta(n+k)$, $k > 0$.

$X_5(z) = z^{k}$, $k > 0$, ROC: entire z-plane except z = ∞.

2 Z-Transform

2.2. Examples

II. Determine the z-transforms of the signal: $x(n) = (\frac{1}{2})^n u(n)$

Solution:

$$x(n) = \{1, (\tfrac{1}{2}), (\tfrac{1}{2})^2, (\tfrac{1}{2})^3, \ldots, (\tfrac{1}{2})^n, \ldots\}$$

Consist an infinite number of nonzero values.

$$X(z) = 1 + \tfrac{1}{2}z^{-1} + (\tfrac{1}{2})^2 z^{-2} + (\tfrac{1}{2})^n z^{-n} + \cdots = \sum_{n=0}^{\infty} (\tfrac{1}{2})^n z^{-n} = \sum_{n=0}^{\infty} (\tfrac{1}{2}z^{-1})^n$$

$$1 + A + A^2 + A^3 + \cdots = \frac{1}{1 - A} \qquad \text{if } |A| < 1$$

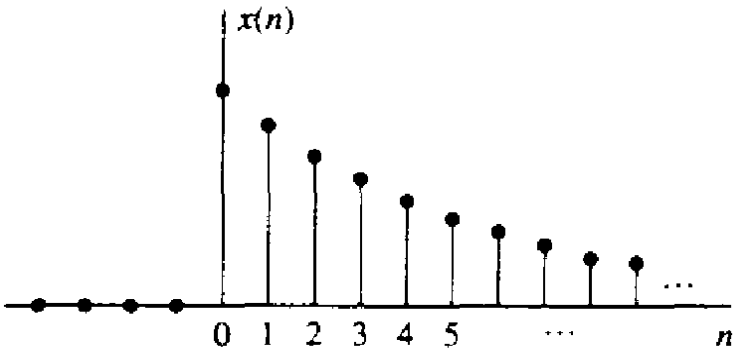
Consequently, for $|\frac{1}{2}z^{-1}| < 1$ **X(z) converges to**

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \qquad \text{ROC: } |z| > \frac{1}{2}$$

2.2. Examples

III. Determine the z-transforms of the signal:

$$x(n) = a^n u(n)$$



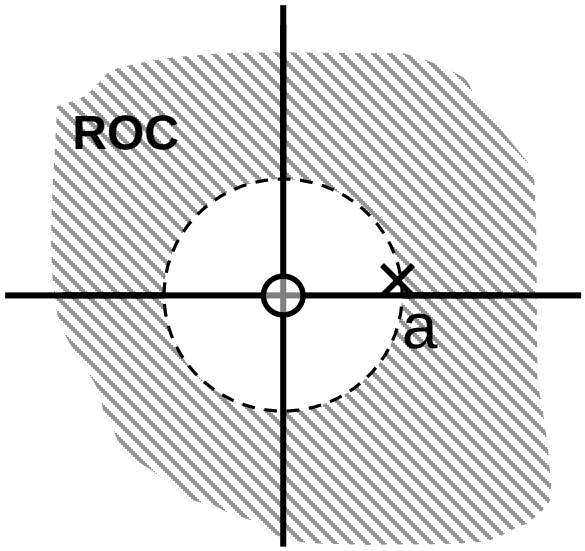
The z-transform is given by:

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

Which converges to:

$$|\alpha z^{-1}| < 1 \text{ or equivalently, } |z| > |\alpha|$$

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \quad , ROC: |z| > |a|$$



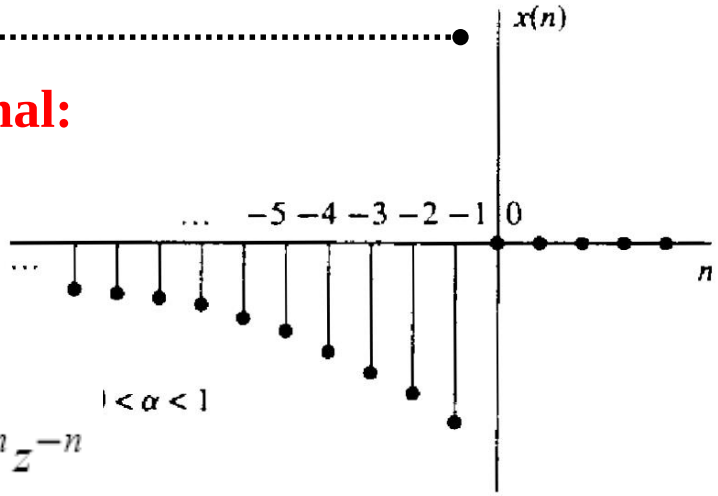
Clearly, *X*(*z*) has a zero at *z* = 0 and a pole at *z* = *a*.

2 Z-Transform

2.2. Examples

IV. Determine the z-transforms of the signal:

$$x(n) = -\alpha^n u(-n-1) = \begin{cases} 0, & n \geq 0 \\ -\alpha^n, & n \leq -1 \end{cases}$$

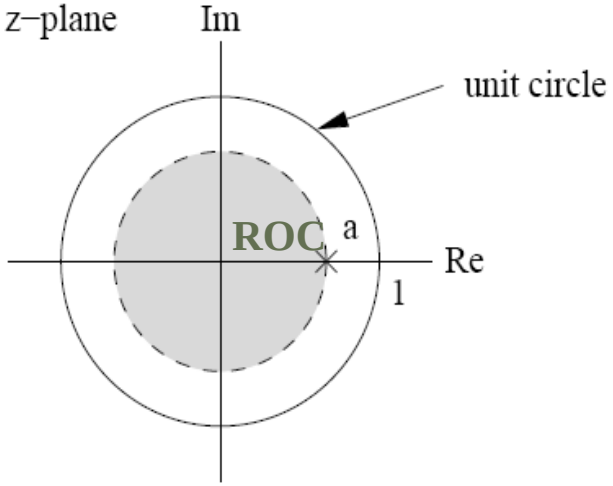


$$X(z) = \sum_{n=-\infty}^{\infty} -a^n u[-n-1] z^{-n} = -\sum_{n=-\infty}^{-1} a^n z^{-n}$$

$$= -\sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1} z)^n.$$

For $|a^{-1}z| < 1$, or $|z| < |a|$, the series converges to

$$X(z) = 1 - \frac{1}{1 - a^{-1}z} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \qquad |z| < |a|.$$

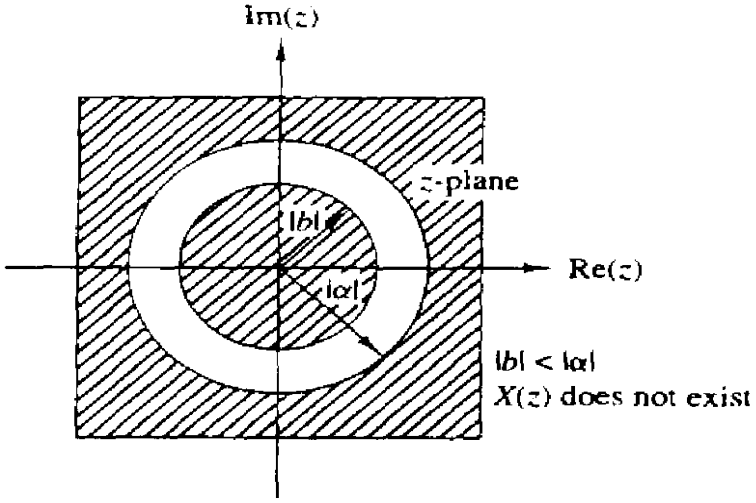


2.2. Examples

V. Determine the z-transforms of the signal:

$$x(n) = \alpha^n u(n) + b^n u(-n-1)$$

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} + \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n + \sum_{l=1}^{\infty} (b^{-1} z)^l$$



- The first power series converges if

$$|\alpha z^{-1}| < 1 \text{ or } |z| > |\alpha|$$

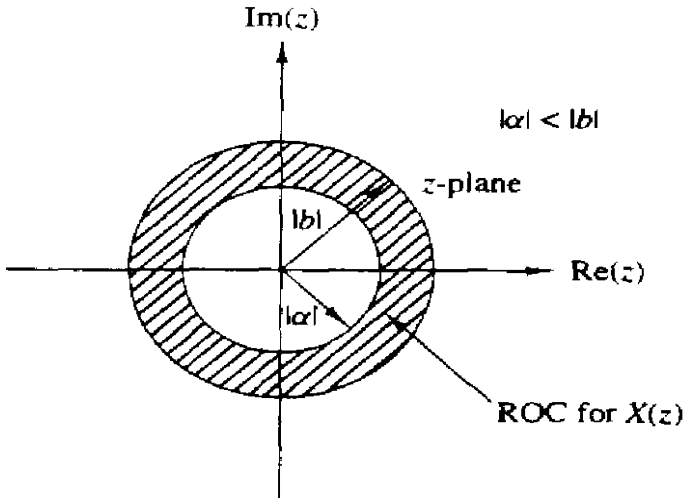
- The second power series converges if

$$|b^{-1} z| < 1 \text{ or } |z| < |b|$$

In determining the convergence of X(z):

Case 1: $|b| < |\alpha|$

Case 2: $|b| > |\alpha|$



Digital Signal Processing

2 Z-Transform

2.3. Z-Transform Properties

Linearity: If $x_1(n) \overset{z}{\leftrightarrow} X_1(z)$ And $x_2(n) \overset{z}{\leftrightarrow} X_2(z)$

Then $x(n) = a_1 x_1(n) + a_2 x_2(n) \overset{z}{\leftrightarrow} a_1 X_1(z) + a_2 X_2(z)$

Example: $x(n) = \left[3(2^n) - 4(3^n)\right] u(n)$

$x_1(n) = 2^n u(n)$ $x_2(n) = 3^n u(n)$ $x(n) = 3x_1(n) - 4x_2(n)$

Its z-transform: $X(z) = 3X_1(z) - 4X_2(z)$

$\alpha^n u(n) \overset{z}{\leftrightarrow} \dfrac{1}{1 - \alpha z^{-1}}$ ROC: $|z| > |\alpha|$

The intersection of the ROC of X1(z) and X2(z) is |z|.3

$X(z) = \dfrac{3}{1 - 2z^{-1}} - \dfrac{4}{1 - 3z^{-1}}$ ROC: $|z| > 3$

2

Time Shifting If $x(n) \overset{z}{\leftrightarrow} X(z)$

Then $x(n-k) \overset{z}{\leftrightarrow} z^{-k}X(z)$

ROC: same as X(z) accept for |z|=0 if k>0 and |z|=∞ if k<0

Example: $x_1(n) = \{\underset{\uparrow}{1}, 2, 5, 7, 0, 1\}$

Its z-transform: $X_1(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$ ROC: $|z| \neq 0$

$x_2(n) = x_1(n+2)$

$X_2(z) = z^2 + 2z^1 + 5 + 7z^{-1} + z^{-3}$ ROC: $|z| \neq \infty$ $|z| \neq 0$

$x_2(n) = x_1(n-2)$

$X_3(z) = z^{-2} + 2z^{-3} + 5z^{-4} + 7z^{-5} + z^{-7}$ ROC: $|z| \neq 0$

2

Scaling in the z-domain

If $x(n) \overset{z}{\leftrightarrow} X(z)$ ROC: $r_1 < |z| < r_2$

Then $a^n x(n) \overset{z}{\leftrightarrow} X(a^{-1}z)$ ROC: $|a|r_1 < |z| < |a|r_2$

for any constant a, real or complex

Example:

$$x(n) = a^n(\cos\omega_0 n)u(n)$$

$$(\cos\omega_0 n)u(n) \overset{z}{\leftrightarrow} \frac{1 - z^{-1}\cos\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}}$$

ROC: $|z| > 1$

Its z-transform:

$$X(z) = \frac{1 - az^{-1}\cos\omega_0}{1 - 2az^{-1}\cos\omega_0 + a^2 z^{-2}}$$

ROC: $|z| > |a|$

2

Time reversal

If $x(n) \overset{z}{\leftrightarrow} X(z)$ ROC: $r_1 < |z| < r_2$

Then $x(-n) \overset{z}{\leftrightarrow} X(z^{-1})$ ROC: $\frac{1}{r_1} < |z| < \frac{1}{r_2}$

Example:

$x(n) = u(-n)$

$u(n) \overset{z}{\leftrightarrow} \frac{1}{1 - z^{-1}}$ ROC: $|z| > 1$

z-transform:

$u(-n) \overset{z}{\leftrightarrow} \frac{1}{1 - z}$ ROC: $|z| < 1$

2

Differentiation in the z-domain If $x(n) \overset{z}{\leftrightarrow} X(z)$ ROC: $r_1 < |z| < r_2$

Then $nx(n) \overset{z}{\leftrightarrow} -z\frac{dX(z)}{dz}$ ROC: $r_1 < |z| < r_2$

Example: $x(n) = na^n u(n)$

$$a^n u(n) \overset{z}{\leftrightarrow} \frac{1}{1-az^{-1}}$$ ROC: $|z| > |a|$

z-transform:

$$na^n u(n) \overset{z}{\leftrightarrow} X(z) = -z\frac{d}{dz}\left(\frac{1}{1-az^{-1}}\right) = \frac{az^{-1}}{(1-az^{-1})^2}$$ ROC: $|z| > |a|$

2

Convolution of two sequences If

$x_1(n) \overset{z}{\leftrightarrow} X_1(z)$ and $x_2(n) \overset{z}{\leftrightarrow} X_2(z)$

Then $x(n) = x_1(n) * x_2(n) \overset{z}{\leftrightarrow} X(z) = X_1(z)X_2(z)$

ROC: is the intersection of $X_1(z)$ and $X_2(z)$ and

Example: $x_1(n) = \{1, -2, 1\}$ $x_2(n) = \begin{cases} 1 & 0 \le n \le 5 \\ 0 & \text{elsewhere} \end{cases}$

$X_1(z) = 1 - 2z^{-1} + z^{-2}$ $X_2(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5}$

z-transform:

$X(z) = X_1(z)X_2(z) = 1 - z^{-1} - z^{-6} + z^{-7}$ ROC: $|z| \neq 0$

$x(n) = \{1, -1, 0, 0, 0, 0, -1, 1\}$

2

Correlation of two sequences:

$x_1(n) \overset{z}{\leftrightarrow} X_1(z)$ and $x_2(n) \overset{z}{\leftrightarrow} X_2(z)$

Then $r_{x_1x_2}(l) = \sum_{n=-\infty}^{\infty} x_1(n)x_2(n-l) \overset{z}{\leftrightarrow} R_{x_1x_2}(z) = X_1(z)X_2(z^{-1})$

Example: The autocorrelation of signal

$x(n) = a^n u(n) \quad -1 < a < 1$

$X(z) = \dfrac{1}{1-az^{-1}}$ ROC: $|z| > |a|$ $X(z^{-1}) = \dfrac{1}{1-az}$ ROC: $|z| > \dfrac{1}{|a|}$

$R_{x_1x_1}(z) = X_1(z)X_1(z^{-1})$ $R_{x_1x_1}(z) = \dfrac{1}{1-az^{-1}}\dfrac{1}{1-az} = \dfrac{1}{1-a(z+z^{-1})+a^2}$

2

The initial Value Theorem:

If x(n) is causal (x(n)=0 for n<0)

$$x(0) = \lim_{z \to \infty} X(z)$$

2

Table 3.2 Some z-transform properties.

	Property	Sequence	Transform	ROC
		$x[n]$	$X(z)$	R_x
		$x_1[n]$	$X_1(z)$	R_{x_1}
		$x_2[n]$	$X_2(z)$	R_{x_2}
1.	Linearity	$a_1x_1[n] + a_2x_2[n]$	$a_1X_1(z) + a_2X_2(z)$	At least $R_{x_1} \bigcap R_{x_2}$
2.	Time shifting	$x[n-k]$	$z^{-k}X(z)$	R_x except $z = 0$ or ∞
3.	Scaling	$a^n x[n]$	$X(a^{-1}z)$	$\lvert a \rvert R_x$
4.	Differentation	$nx[n]$	$-z\dfrac{dX(z)}{dz}$	R_x
5.	Conjugation	$x^*[n]$	$X^*(z^*)$	R_x
6.	Real-part	$\mathrm{Re}\{x[n]\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	At least R_x
7.	Imaginary part	$\mathrm{Im}\{x[n]\}$	$\frac{1}{2}[X(z) - X^*(z^*)]$	At least R_x
8.	Folding	$x[-n]$	$X(1/z)$	$1/R_x$
9.	Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	At least $R_{x_1} \bigcap R_{x_2}$
10.	Initial-value theorem	$x[n] = 0$ for $n < 0$	$x[0] = \lim_{z\to\infty} X(z)$	

2.3 Z-Transform Properties

Examples:

I. Determine X(z).

$$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-az^{-1}}, \qquad |z| > a$$

$$x[n] = na^n u[n]$$

$$X(z) = -z\frac{d}{dz}\left(\frac{1}{1-az^{-1}}\right) = \frac{az^{-1}}{(1-az^{-1})^2}, \qquad |z| > a$$

2.3. Z-Transform Properties

Examples:

II. Determine X(z).

$$x[n] = a^{-n}u[-n]$$

This signal is time reversed of

$$a^{n}u[n].$$

The z-transform is

$$X(z) = \frac{1}{1-az} = \frac{-a^{-1}z^{-1}}{1-a^{-1}z^{-1}}, \qquad |z| < |a^{-1}|$$

2 Rational Z-Transforms

Rational z-transform plays an important role in DSP,in particular, in studying IIR filters.

are those for which X(z) is a rational function, that is a ratio of two polynomials.

Poles and zeros

Pole-zero characterizations are critical in determining realizable &stable systems.

The zeros of a z-transform X(z) are values of z for which X(z)=0.

The poles of a z-transform are the values of z for which X(z)= ∞ ,if X(z) is rational function,

then

$$X(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z + b_2 z^{-2} + \cdots + b_M z^{-M}}{a_0 + a_1 z + a_2 z^{-2} + \cdots + a_N z^{-N}}$$

2

If $a_0 \neq 0$ and $b_0 \neq 0$, we can avoid the negative powers of z by factoring out the terms $b_0 z^{-M}$ and $a_0 z^{-N}$ as follows:

$$H(z) = \frac{b_0 z^{-M}}{z^{-N}} \times \frac{z^M + (b_1/b_0)z^{M-1} + \cdots + (b_M/b_0)}{z^N - a_1 z^{N-1} - \cdots - a_N}$$

$$H(z) = b_0\, z^{-M+N} \times \frac{(z - z_1)(z - z_2)\cdots(z - z_M)}{(z - p_1)(z - p_2)\cdots(z - p_N)}$$

$$H(z) = G\, z^{-M+N} \times \frac{\prod_{k=1}^{M}(z - z_k)}{\prod_{k=1}^{N}(z - p_k)}$$

where $G \equiv b_0/a_0$. Thus $X(z)$ has M finite zeros at $z = z_1, z_2, \ldots, z_M$ (the roots of the numerator polynomial), N finite poles at $z = p_1, p_2, \ldots, p_N$ (the roots of the denominator polynomial), and $|N - M|$ zeros (if $N > M$) or poles (if $N < M$) at the origin $z = 0$.

2

Example: determine the pole –zero plot for the signal. $X(n)=a^n u(n)$ $a>0$

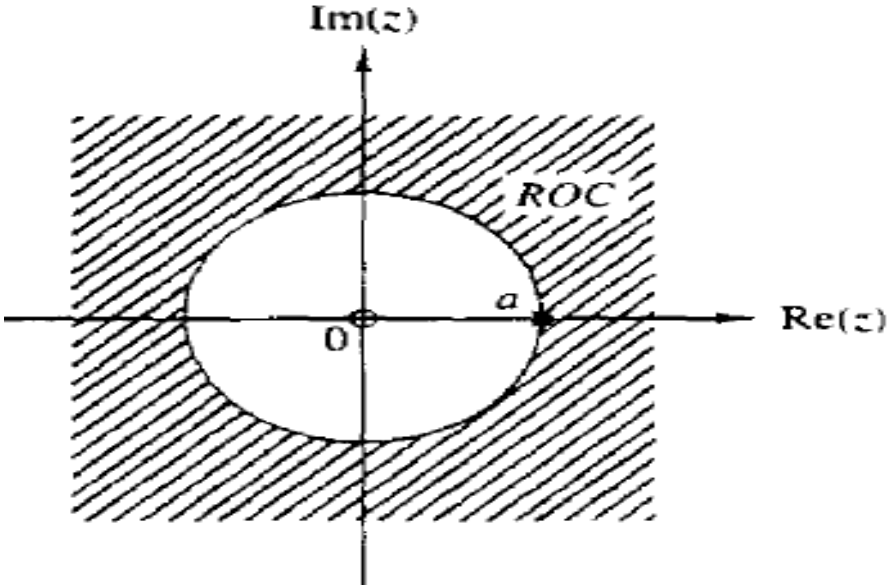
from the table ,

Thus x(z) has one zero at $z_1=0$ anc included in the ROC since the z-t

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \qquad \text{ROC: } |z| > a$$

the pole –zero plot is shown in fig. below

2



Pole –zero plot for the causal exponential signal.

Example: determine the pole-zero plot for signal

Where a>0.

$$x(n) = \begin{cases} a^n, & 0 \le n \le M-1 \\ 0, & \text{elsewhere} \end{cases}$$

2

From the definition,we obtain

Since a>0,the ec $$X(z) = \sum_{n=0}^{M-1} (az^{-1})^n = \frac{1-(az^{-1})^M}{1-az^{-1}} = \frac{z^M - a^M}{z^{M-1}(z-a)}$$

The zero z_0 =a cancels the pole at z=a.thus

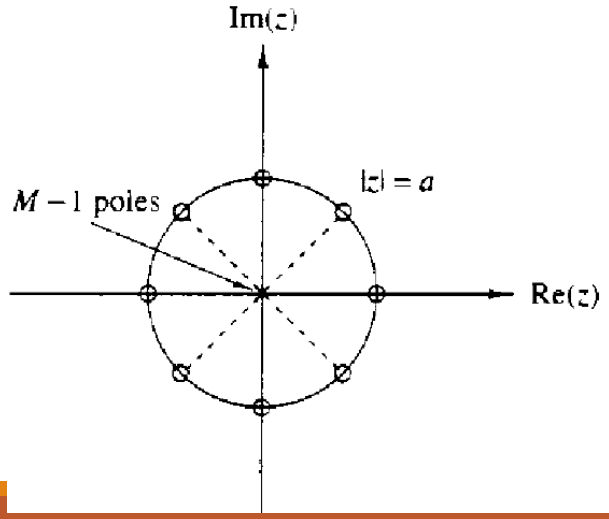
$$z_k = ae^{j2\pi k/M} \qquad k = 0, 1, \ldots, M-1$$

$$X(z) = \frac{(z-z_1)(z-z_2)\cdots(z-z_{M-1})}{z^{M-1}}$$

2

which has M-1 zeros and M-1 poles,located in figure below.

Note that: the ROC is the entire z-plane except z=0 because of the M-1 poles located at the origin.



Pole-zero pattern for the finite duration signal

$x(n)=a^{n.}$

$$0 \leq n \leq M-1 (a>0).$$

2 2.4 The Inverse Z-Transform

The procedure of transforming from the z-domain to the time domain is called inverse z-transform.

x(n) can be obtain from X(z) using the Cauchy integral theorem. Multiplying both side z^{n-1} and integral both sides over a closed contour with in ROC of X(z) which encloses the origin.

$$\oint_C X(z)z^{n-1}dz = \oint_C \sum_{k=-\infty}^{\infty} x(k)z^{n-1-k}dz$$

$$\oint_C X(z)z^{n-1}dz = \sum_{k=-\infty}^{\infty} x(k)\oint_C z^{n-1-k}dz$$

The Cauchy integral theorem, which states that

$$\frac{1}{2\pi j}\oint_C z^{n-1-k}dz = \begin{cases} 1 & k=n \\ 0 & k\neq n \end{cases}$$

Applying this the previous equation, we have inversion formula

$$x(n) = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz$$



Z-Transform

2.4 The Inverse Z-Transform

The inverse z-transform formally given by

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

Here the symbol $\oint$ indicates an integration in counterclockwise direction around a closed path in the complex z-plane (known as contour integral).

Such contour integral is difficult to evaluate (but could be done using Cauchy's residue theorem), therefore we often use other techniques to obtain the inverse z-tranform.

2

there are three methods that are used for the evaluation of the inverse z-transform:

Inspection method

Expansion into a power series of terms

Partial-fraction expansion and table lookup

2

2.4.1. Inspection Method:

You can find the inverse z-transform from z-transform pair the table.

Example:

Using the table

$$X(z) = \left(\frac{1}{1 - \frac{1}{2}z^{-1}} \right), \qquad |z| > \frac{1}{2}$$

Determine x(n).

$$a^n u[n] \overset{Z}{\longleftrightarrow} \frac{1}{1 - az^{-1}}, \qquad \text{for } |z| > |a|$$

then

$$x[n] = \left(\frac{1}{2} \right)^n u[n]$$

2.4. The Inverse Z-Transform

2.4.2. Power Series Expansion:

Given a z-transform x(z) with its corresponding ROC ,we can expand X(z) into power series.

If the z-transform is given as power series in the form:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \ldots + x[-2]z^{2} + x[-1]z^{1} + x[0] + x[1]z^{-1} + x[2]z^{-2} + \ldots$$

Which converges in the given ROC .when X(z) is a rational, the expansion can be performed by long division.

x(n) can be found by identifying the coefficient of the appropriate power of z^{-1}.

2.4. The Inverse Z-Transform

2.4.2. Power Series Expansion:

Example:

$$X(z) = z^2(1 - \frac{1}{2}z^{-1})(1 + z^{-1})(1 - z^{-1}) \qquad X(z) = z^2 - \frac{1}{2}z - 1 + \frac{1}{2}z^{-1}$$

By inspection,

$$x[n] = \begin{cases} 1 & n = -2 \\ -\frac{1}{2} & n = -1 \\ -1 & n = 0 \\ \frac{1}{2} & n = 1 \\ 0 & \text{otherwise} \end{cases}$$

or

$$x[n] = 1\delta[n+2] - \frac{1}{2}\delta[n+1] - 1\delta[n] + \frac{1}{2}\delta[n-1]$$

2

Example: determine the inverse z-transform of

When $$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

a)ROC: |z|>1

b)ROC: |z|< 0.5

a) Since the ROC is the exterior of the circle,we expect x(n) to be a causal signal. thus we seek a power series expansion in negative powers of z.

2

By dividing numerator of X(z) by its denominator we obtain the power series

$$X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \frac{31}{16}z^{-4} + \cdots$$

$$x(n) = \{1, \underset{\uparrow}{\frac{3}{2}}, \frac{7}{4}, \frac{15}{8}, \frac{31}{16}, \ldots\}$$

Each step of the long division process, eliminate the lowest power term of z^{-1}

2

b) In this case the ROC is the interior of a circle. Consequently the signal x(n) is anticausal.to obtain a power series expansion in positive powers of z. we perform long division in the following way:

$$
\begin{array}{r}
2z^{2}+6z^{3}+14z^{4}+30z^{5}+62z^{6}+\cdots \\
\hline
\tfrac{1}{5}z^{-2}-\tfrac{3}{5}z^{-1}+1\;\big)\;1 \\
1-3z+2z^{2} \\
\hline
3z-2z^{2} \\
3z-9z^{2}+6z^{3} \\
\hline
7z^{2}-6z^{3} \\
7z^{2}-21z^{3}+14z^{4} \\
\hline
15z^{3}-14z^{4} \\
15z^{3}-45z^{4}+30z^{5} \\
\hline
31z^{4}-30z^{5}
\end{array}
$$

2

thus

$$X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = 2z^2 + 6z^3 + 14z^4 + 30z^5 + 62z^6 + \cdots$$

In this case x(n)=0 for n>=0.we conclude that

In each step of the long division process,the lowest term of z is eliminated.

$$x(n) = \{\cdots 62, 30, 14, 6, 2, 0, \underset{\uparrow}{0}\}$$

2 Z-Transform

2.4. The Inverse Z-Transform

2.4.3. Partial Fraction Expansion:

For any rational function we can obtain a partial fraction expansion, and identify the z-transform for each term.

$$X(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

$$X(z) = \frac{b_0}{a_0} \frac{\prod_{k=1}^{M}(1 - c_k z^{-1})}{\prod_{k=1}^{N}(1 - d_k z^{-1})}$$

The most general form for partial fraction expansion, which can deal with multiple order poles is:

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1, k\neq i}^{N} \frac{A_k}{1 - d_k z^{-1}} + \sum_{m=1}^{s} \frac{C_m}{(1 - d_i z^{-1})^m}$$

If x(z) has a second order pole for example:

$$\frac{B_1}{1 - \alpha_k z^{-1}} + \frac{B_2}{(1 - \alpha_k z^{-1})^2}$$

$$B_1 = \alpha_k \left[\frac{d}{dz} (1 - \alpha_k z^{-1})^2 X(z) \right]_{z=\alpha_k}$$

$$B_2 = \left[(1 - \alpha_k z^{-1})^2 X(z) \right]_{z=\alpha_k}$$



- If we have a rational function with *distinct* poles

$$X(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots + b_M z^{-M}}{1 + a_1 z^{-1} + a_2 z^{-2} + \ldots a_N z^{-N}}$$

the complete partial fraction expansion takes the form

$$X(z) = \sum_{k=0}^{M-N} C_k z^{-k} + \sum_{k=1}^{N} \frac{A_k}{(1 - p_k z^{-1})}$$

with $A_k = (1 - p_k z^{-1}) X(z)\Big|_{z=p_k}$.

- Thus, for example, if $x[n]$ is known to be causal, then

$$x[n] = \sum_{k=0}^{M-N} C_k \delta[n-k] + \sum_{k=1}^{N} A_k (p_k)^n$$

with $\mathrm{ROC} : |z| > \max_{1 \le k \le N}\{|p_k|\}$.

- The parameters $\{C_k, p_k, A_k\}$ can be computed using the MATLAB function `[A,p,C]=residuez(b,a)`.

2 Z-Transform

2.4. The Inverse Z-Transform

2.4.3. Partial Fraction Expansion:

Example:

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{(1 + z^{-1})^2}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}, \qquad |z| > 1$$

Determine the x(n).

M = N = 2, so this can be expressed as

$$X(z) = B_0 + \frac{A_1}{1 - \frac{1}{2}z^{-1}} + \frac{A_2}{1 - z^{-1}}$$

The value B_0 can be found by long division:

$$\begin{array}{r} 2 \\ \frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 \overline{)\; z^{-2} + 2z^{-1} + 1} \\ \underline{z^{-2} - 3z^{-1} + 2} \\ 5z^{-1} - 1 \end{array}$$

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})}$$

so

2.4. The Inverse Z-Transform

2.4.2. Partial Fraction Expansion:

Example: (cont.)

$$A_k = (1 - d_k z^{-1}) X(z)\Big|_{z=d_k}$$

The coefficients A_1 and A_2 can be found using:

So

$$A_1 = \left.\frac{1+2z^{-1}+z^{-2}}{1-z^{-1}}\right|_{z^{-1}=2} = \frac{1+4+4}{1-2} = -9$$

and

$$A_2 = \left.\frac{1+2z^{-1}+z^{-2}}{1-\frac{1}{2}z^{-1}}\right|_{z^{-1}=1} = \frac{1+2+1}{1/2} = 8$$

Therefore,

$$X(z) = 2 - \frac{9}{1-\frac{1}{2}z^{-1}} + \frac{8}{1-z^{-1}}$$

Using $\text{ROC is } |z| > 1$ **and inspection method,**

then

$$x[n] = 2\delta[n] - 9(1/2)^n u[n] + 8u[n]$$



Example

Consider a sequence $x[n]$ with z-transform

$$X(z) = \frac{(1+z^{-1})}{(1-z^{-1})(1-0.5z^{-1})}$$

which is a *proper* rational function because the degree of the numerator is less than the degree of the denominator.

$$\frac{(1+z^{-1})}{(1-z^{-1})(1-0.5z^{-1})} = \frac{A_1}{(1-z^{-1})} + \frac{A_2}{(1-0.5z^{-1})} = \frac{A_1(1-0.5z^{-1}) + A_2(1-z^{-1})}{(1-z^{-1})(1-0.5z^{-1})}$$

Therefore,

$$z + 1 = A_1(z-0.5) + A_2(z-1)$$

which (by equating the coefficients of the same degree polynomial terms) leads to

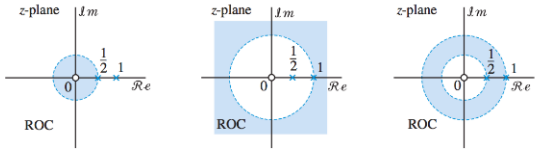
$$\begin{aligned} A_1 + A_2 &= 1 \\ -A_1/2 - A_2 &= 1 \end{aligned}$$

Solving the system results in $A_1 = 4$ and $A_2 = -3$, and hence

$$X(z) = \frac{4}{1-z^{-1}} - \frac{3}{1-0.5z^{-1}}$$



There are three choices of the ROC for $X(z)$:



- **Choice 1:** ROC : $|z| < 0.5$

$$x[n] = -4 \cdot u[-n-1] + 3 \cdot 0.5^n u[-n-1] \quad \text{(anticausal)}$$

- **Choice 2:** ROC : $|z| > 1$

$$x[n] = 4 \cdot u[n] - 3 \cdot 0.5^n u[n] \quad \text{(causal)}$$

- **Choice 3:** ROC : $0.5 < |z| < 1$

$$x[n] = -4 \cdot u[-n-1] - 3 \cdot 0.5^n u[n] \quad \text{(two-sided)}$$

2 Find inverse z-transform – real unique poles

- Find the inverse z-transform of:

$$X[z] = \frac{8z - 19}{(z-2)(z-3)}$$

- Step 1: Divide both sides by z:

$$\frac{X[z]}{z} = \frac{8z - 19}{z(z-2)(z-3)}$$

- Step 2: Perform partial fraction:

$$\frac{X[z]}{z} = \frac{(-19/6)}{z} + \frac{(3/2)}{z-2} + \frac{(5/3)}{z-3}$$

- Step 3: Multiply both sides by z:

$$X[z] = -\frac{19}{6} + \frac{3}{2}\left(\frac{z}{z-2}\right) + \frac{5}{3}\left(\frac{z}{z-3}\right)$$

- Step 4: Obtain inverse z-transform of each term from table

$$x[n] = -\tfrac{19}{6}\delta[n] + \left[\tfrac{3}{2}(2)^n + \tfrac{5}{3}(3)^n\right] u[n]$$

2 Find inverse z-transform – repeat real poles (1)

- Find the inverse z-transform of: $X[z] = \frac{z(2z^2 - 11z + 12)}{(z-1)(z-2)^3}$
- Divide both sides by z and expand:

$$\frac{X[z]}{z} = \frac{2z^2 - 11z + 12}{(z-1)(z-2)^3} = \frac{k}{z-1} + \frac{a_0}{(z-2)^3} + \frac{a_1}{(z-2)^2} + \frac{a_2}{(z-2)}$$

- Use covering method to find k and a_0:

$$k = \left.\frac{2z^2 - 11z + 12}{(z-1)(z-2)^3}\right|_{z=1} = -3 \qquad a_0 = \left.\frac{2z^2 - 11z + 12}{(z-1)(z-2)^3}\right|_{z=2} = -2$$

- We get:

$$\frac{X[z]}{z} = \frac{2z^2 - 11z + 12}{(z-1)(z-2)^3} = \frac{-3}{z-1} - \frac{2}{(z-2)^3} + \frac{a_1}{(z-2)^2} + \frac{a_2}{(z-2)}$$

- To find a_2, multiply both sides by z and let $z \to \infty$:

$$0 = -3 - 0 + 0 + a_2 \implies a_2 = 3$$

2

- To find a_1, let z = 0:

$$\frac{12}{8} = 3 + \frac{1}{4} + \frac{a_1}{4} - \frac{3}{2} \quad \Longrightarrow \quad a_1 = -1$$

- Therefore, we find:

$$\frac{X[z]}{z} = \frac{-3}{z-1} - \frac{2}{(z-2)^3} - \frac{1}{(z-2)^2} + \frac{3}{z-2}$$

$$X[z] = -3\frac{z}{z-1} - 2\frac{z}{(z-2)^3} - \frac{z}{(z-2)^2} + 3\frac{z}{z-2}$$

- Use pairs #6 & #10

6	$\gamma^n u[n]$	$\Longleftrightarrow \quad \frac{z}{z-\gamma}$
10	$\frac{n(n-1)(n-2)\cdots(n-m+1)}{\gamma^m m!}\gamma^n u[n]$	$\Longleftrightarrow \frac{z}{(z-\gamma)^{m+1}}$

$$x[n] = \left[-3 - 2\frac{n(n-1)}{8}(2)^n - \frac{n}{2}(2)^n + 3(2)^n\right] u[n]$$

$$= -\left[3 + \frac{1}{4}(n^2 + n - 12)2^n\right] u[n]$$

$$a^n u[n] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1 - az^{-1}}, \qquad |z| > a$$

2 Find inverse z-transform – complex poles

- Find inverse z-tranform of: $$X[z] = \frac{2z(3z+17)}{(z-1)(z^2-6z+25)} = \frac{2z(3z+17)}{(z-1)(z-3-j4)(z-3+j4)}$$
- Whenever we encounter complex pole, we need to use a special partial fraction method (called quadratic factors):

$$\frac{X[z]}{z} = \frac{2(3z+17)}{(z-1)(z^2-6z+25)} = \frac{2}{z-1} + \frac{Az+B}{z^2-6z+25}$$

- Now multiply both sides by z, and let $z \to \infty$:

$$0 = 2 + A \Longrightarrow A = -2$$

- We get:

$$\frac{2(3z+17)}{(z-1)(z^2-6z+25)} = \frac{2}{z-1} + \frac{-2z+B}{z^2-6z+25}$$

2 Z-Transform

2.5. Transfer Function

The output y(n) of LTI system to an input sequence x(n) can be obtained by computing the convolution of x(n) with the unit sample (Impulse) response h(n) of the system.

$$y(n) = h(n) * x(n)$$

Expressing this relationship in the z-domain as ,

$$H(z) = \frac{Y(z)}{X(z)}$$

$$Y(z) = H(z)\,X(z)$$

Impulse Response:

$$H(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n}$$

2.5. Transfer Function LTI systems

The LTI system can be described by means of a constant coefficient linear difference equation as follows:

$$y(n) = \sum_{k=0}^{N} b(k)x(n-k) - \sum_{k=1}^{M} a(k)y(n-k)$$

$$Y(z) = \sum_{k=0}^{N} b(k)z^{-k}X(z) - \sum_{k=1}^{M} a(k)z^{-k}Y(z)$$

The transfer function of the LTI system:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{N} b(k)z^{-k}}{1+\sum_{k=1}^{M} a(k)z^{-k}}$$

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Consider an LTID system whose input–output relationship is given by the following difference equation:

$$y[k+n] + a_{n-1}y[k+n-1] + \cdots + a_0 y[k] = b_m x[k+m] + b_{m-1}x[k+m-1] + \cdots + b_0 x[k].$$

By taking the z-transform of both sides of the above equation, we obtain

$$\{z^n + a_{n-1}z^{n-1} + \cdots + a_0 z\}Y(z) = \{b_m z^m + b_{m-1}z^{m-1} + \cdots + b_0\}X(z),$$

which reduces to the following transfer function:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_m z^m + b_{m-1}z^{m-1} + \cdots + b_0}{z^n + a_{n-1}z^{n-1} + \cdots + a_0}$$

or alternatively as

$$H(z) = z^{m-n}\frac{b_m + b_{m-1}z^{-1} + \cdots + b_0 z^{-m}}{1 + a_{n-1}z^{-1} + \cdots + a_0 z^{-n}}.$$

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Example

The input–output relationship of an LTID system is given by the following difference equation:

$$y[k+2] - \frac{3}{4}y[k+1] + \frac{1}{8}y[k] = 2x[k+2].$$

Determine the transfer function and the impulse response of the system.

Solution

Substituting $m = k + 2$, Eq. can be written as follows:

$$y[m] - \frac{3}{4}y[m-1] + \frac{1}{8}y[m] = 2x[m].$$

Calculating the z-transform on both sides of the equation yields

$$Y(z) - \frac{3}{4}z^{-1}Y(z) + \frac{1}{8}z^{-2}Y(z) = 2X(z),$$

which results in the following transfer function:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}.$$

2

To calculate the impulse response of the LTID system, consider the partial fraction expansion of $H(z)$ as

$$H(z) = \frac{2}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)} \equiv \frac{4}{1 - \frac{1}{2}z^{-1}} - \frac{2}{1 - \frac{1}{4}z^{-1}}.$$

By calculating the inverse z-transform of both sides, the impulse response $h[k]$ is obtained:

$$h[k] = 4\left(\frac{1}{2}\right)^k u[k] - 2\left(\frac{1}{4}\right)^k u[k],$$

2 Z-Transform

2.5. Transfer Function

Example:

Determine the transfer function and the impulse response of the following system:

$$y(n) = \frac{1}{2}y(n-1) + 2x(n)$$

z-transform

$$Y(z) = \tfrac{1}{2}z^{-1}Y(z) + 2X(z)$$

Transfer Function

$$\frac{Y(z)}{X(z)} \equiv H(z) = \frac{2}{1 - \frac{1}{2}z^{-1}}$$

The inverse transform

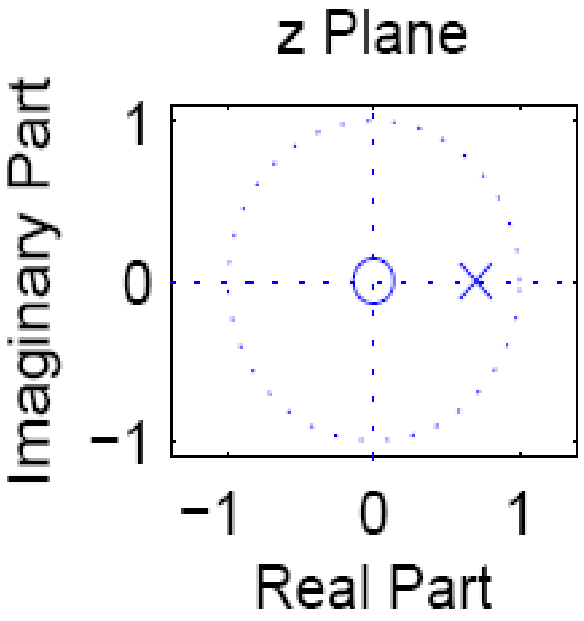
$$h(n) = 2(\tfrac{1}{2})^n u(n)$$

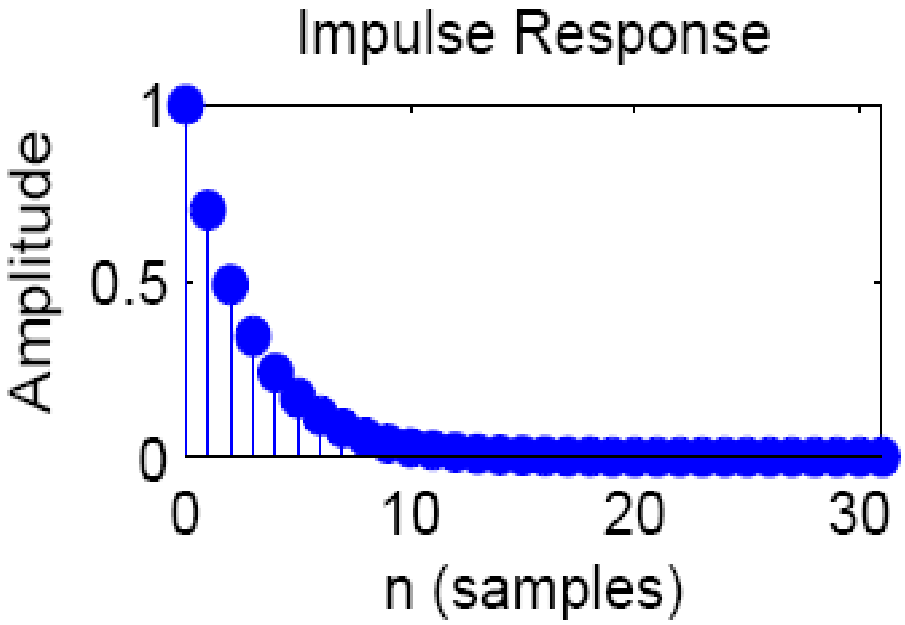
2 Z-Transform

2.5. Transfer Function

Example:

$$H(z) = \frac{z}{z-0.7} = \frac{1}{1-0.7 \cdot z^{-1}}$$

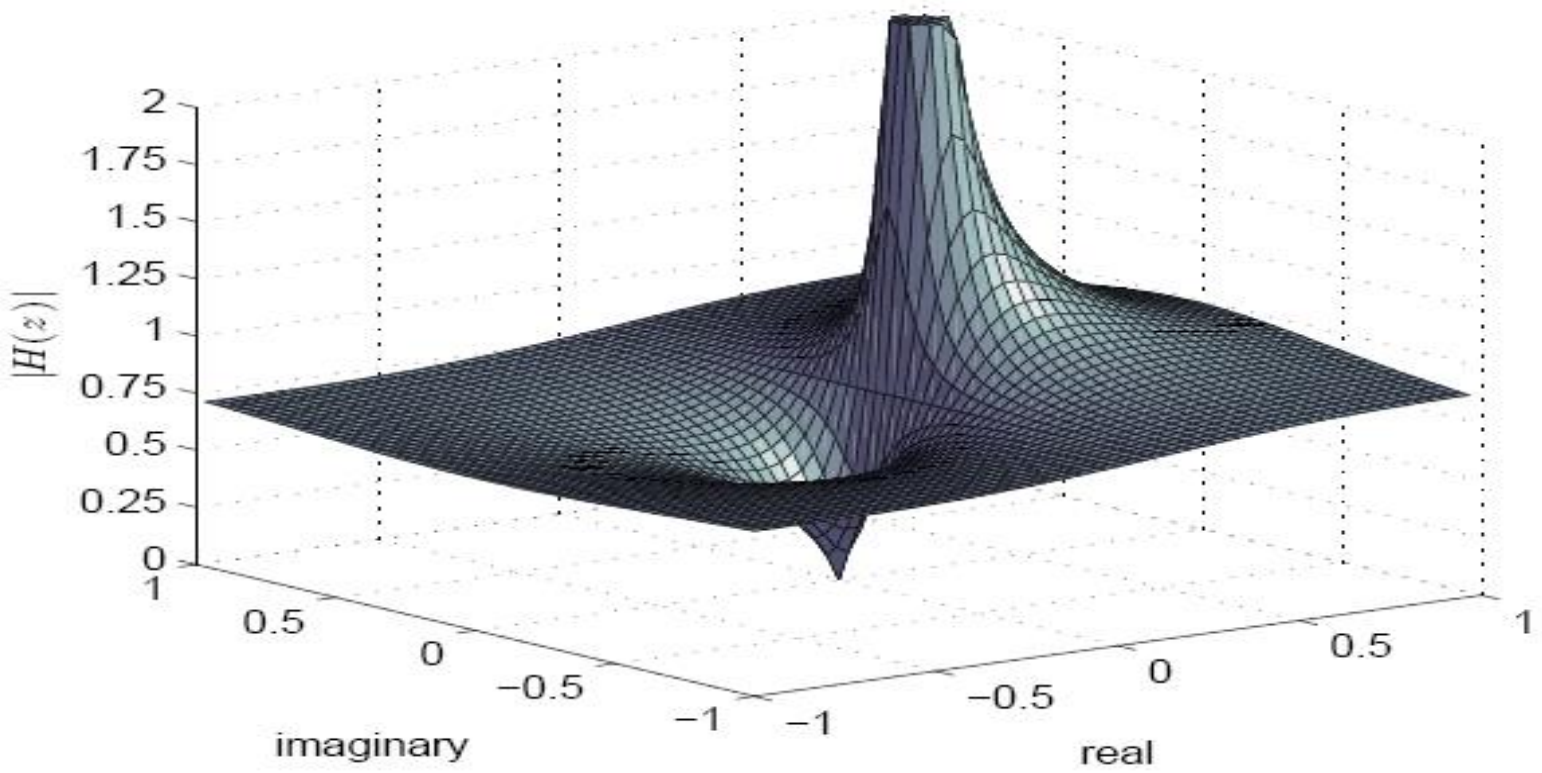




2.5. Transfer Function

Example:

$$H(z) = \frac{0.8}{1 - 0.2 \cdot z^{-1}} = \frac{0.8z}{z - 0.2}$$



2 2.6 Solution of difference equations

An important application of the z-transform is to solve linear, constant-coefficient difference equations. we used a time-domain approach to obtain the zero-input, zero-state, and overall solutions of difference equations.

In this section, we discuss an alternative approach based on the z-transform.

Example

A causal system is represented by the following difference equation:

$$y[k+2] - 5y[k+1] + 6y[k] = 3x[k+1] + 5x[k].$$

Calculate the output $y[k]$ for the input $x[k] = 2^{-k}u[k]$ and the initial conditions $y[-1] = 11/6$, $y[-2] = 37/36$.

Solution

Substituting $k-2$ for k in , we obtain

$$y[k] - 5y[k-1] + 6y[k-2] = 3x[k-1] + 5x[k-2].$$

Note that the input sequence $x[k] = 2^{-k}u[k]$ is causal, hence $x[-2] = x[-1] = 0$. Using the time-shifting property,

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$$3x[k-1]+5x[k-2] \overset{z}{\longleftrightarrow} 3z^{-1}X(z)+5z^{-2}X(z).$$

Using the z-transform pair,

$$x[k]=2^{-k}u[k]=0.5^{k}u[k] \overset{z}{\longleftrightarrow} X(z)=\frac{1}{1-0.5z^{-1}},$$

the z-transform of the right-hand side of Eq. is given by

$$3x[k-1]+5x[k-2] \overset{z}{\longleftrightarrow} \frac{3z^{-1}}{1-0.5z^{-1}}+\frac{5z^{-2}}{1-0.5z^{-1}}=\frac{3z^{-1}+5z^{-1}}{1-0.5z^{-1}}.$$

the z-transform of the left-hand side of Eq. is given by

$$y[k]-5y[k-1]+6y[k-2] \overset{z}{\longleftrightarrow} (1-5z^{-1}+6z^{-2})Y(z)+(11z^{-1}-3).$$

Equating the z-transforms of both sides of Eq. , we obtain

$$(1-5z^{-1}+6z^{-2})Y(z)+(11z^{-1}-3)=\frac{3z^{-1}+5z^{-2}}{1-0.5z^{-1}},$$

which reduces to

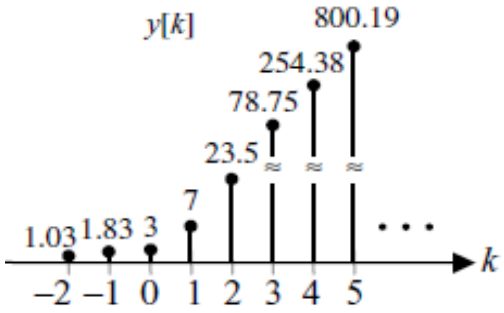
$$Y(z)=\frac{3-11z^{-1}}{1-5z^{-1}+6z^{-2}}+\frac{3z^{-1}+5z^{-2}}{(1-0.5z^{-1})(1-5z^{-1}+6z^{-2})}$$

2

$$= \frac{(3 - 11z^{-1})(1 - 0.5z^{-1}) + 3z^{-1} + 5z^{-2}}{(1 - 0.5z^{-1})(1 - 5z^{-1} + 6z^{-2})}$$
$$= \frac{3 - 9.5z^{-1} + 10.5^{-2}}{(1 - 0.5z^{-1})(1 - 2z^{-1})(1 - 3z^{-1})}.$$

Using partial fraction expansion, $Y(z)$ can be expressed as follows:

$$Y(z) = \frac{26}{15} \times \frac{1}{1 - 0.5z^{-1}} - \frac{7}{3} \times \frac{1}{1 - 2z^{-1}} + \frac{18}{5} \times \frac{1}{1 - 3z^{-1}}.$$



Taking the inverse transform, we obtain

$$y[k] = \left[\frac{26}{15} \times 0.5^k - \frac{7}{3} \times 2^k + \frac{26}{15} \times 3^k\right] \text{ for } k > 0.$$

The output response is plotted in Fig.



Z-Transform

2.7. Poles, Zeros, and Pole-Zero Plot

$$H(z)=\frac{\sum_{k=0}^{N} b(k)z^{-k}}{1+\sum_{k=1}^{M} a(k)z^{-k}}=\frac{b_0}{a_0}z^{N-M}\frac{\prod_{k=1}^{N}(z-z_k)}{\prod_{k=1}^{M}(z-p_k)}$$

$H(p_k)\to\infty$

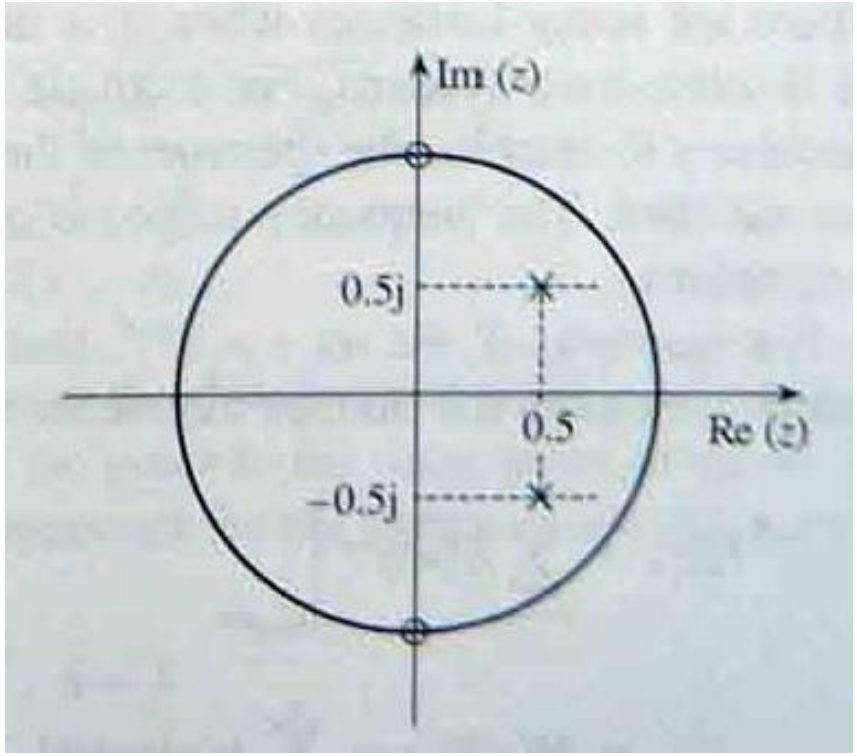
Zeros of *H(z)*: the set $\{z_k\}$ of *z*-plane for which $H(z_k)=0$

Poles of *H(z)*: the set $\{p_k\}$ of *z* -plane for which

Pole-zero plot: the plot of the zeros and the poles of *H(z)* in the z-plane represents a strong tool for LTI system description.

2.7. Poles, Zeros, and Pole-Zero Plot

$$H(z) = \frac{(z-2)(z+1)z}{(z-0.5+j0.5)(z-0.5-j0.5)(z-0.75)}$$



2

Example

Determine the poles and zeros of the following LTID systems:

$$H_1(z) = \frac{z}{z^2 - 3z + 2};$$

$$H_2(z) = \frac{1}{(z - 0.1)(z - 0.5)(z + 0.2)};$$

$$H_3(z) = \frac{z^2(2z - 1.5)}{(z + 0.4)(z - 0.5)^2};$$

$$H_4(z) = \frac{z^2 + 0.7z + 1.6}{(z^2 - 1.2z + 1)(z + 0.3)}.$$

Solution

(i) $H_1(z) = \dfrac{z}{z^2 - 3z + 2} = \dfrac{z}{(z - 1)(z - 2)}.$

There is one zero, at $z = 0$, and two poles, at $z = 1$ and 2.

(ii) $H_2(z) = \dfrac{1}{(z - 0.1)(z - 0.5)(z + 0.2)}.$

There is no zero, but there are three poles, at $z = 0.1, 0.5$, and -0.2.



(iii) $H_3(z) = \dfrac{z^2(2z - 1.5)}{(z + 0.4)(z - 0.5)^2}$.

There are three zeros, at $z = 0$, 0, and 0.75. There are three poles, at $z = -0.4$, 0.5, and 0.5.

(iv) $H_4(z) = \dfrac{(z - 0.5)(z + 1.2)}{((z - 0.6)^2 + 0.8^2)(z + 0.3)}$

$= \dfrac{(z - 0.5)(z + 1.2)}{(z - 0.6 + \text{j}0.8)(z - 0.6 - \text{j}0.8)(z + 0.3)}$.

There are two zeros, at $z = 0.5$ and -1.2. There are three poles, at $z = 0.6 - \text{j}0.8$, $0.6 + \text{j}0.8$, and -0.3.

2

The Z-Transform

2.8. Causality and Stability

Causality:

- $H(z) = \sum_{n=0}^{\infty} h[n]z^{-n}$ cle and extend to infinity.
- ROC outside the outermost pole and the denominator greater than of numerator.

$$h(n) = 0 \qquad n < 0$$

Causal:

$$H(z) = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-2z^{-1}}, \quad |z| > 2$$

$$= \frac{2z^2 - \frac{5}{2}z}{z^2 - \frac{5}{2}z + 1}$$

$$h[n] = \left[\left(\frac{1}{2}\right)^n + 2^n\right]u[n].$$

Non-causal:

$$H(z) = \frac{z^3 - 2z^2 + z}{z^2 + \frac{1}{4}z + \frac{1}{8}}$$

2.8. Causality and Stability

Stability:

- The ROC includes the unit circle of |z| = 1.
- Rational *H(z)* has poles inside the unit circle.

Stable System Example:

$$H(z)=\frac{1-0.9z^{-1}+0.18z^{-2}}{1-0.8z^{-1}+0.64z^{-2}}$$

$$z_1=0.3 \quad p_1=0.4000+0.6928j \quad |p_1|=0.8<1$$

$$z_2=0.6 \quad p_2=0.4000-0.6928j \quad |p_2|=0.8<1$$

Unstable System Example:

$$H(z)=\frac{1-0.16z^{-2}}{1-1.1z^{-1}+1.21z^{-2}}$$

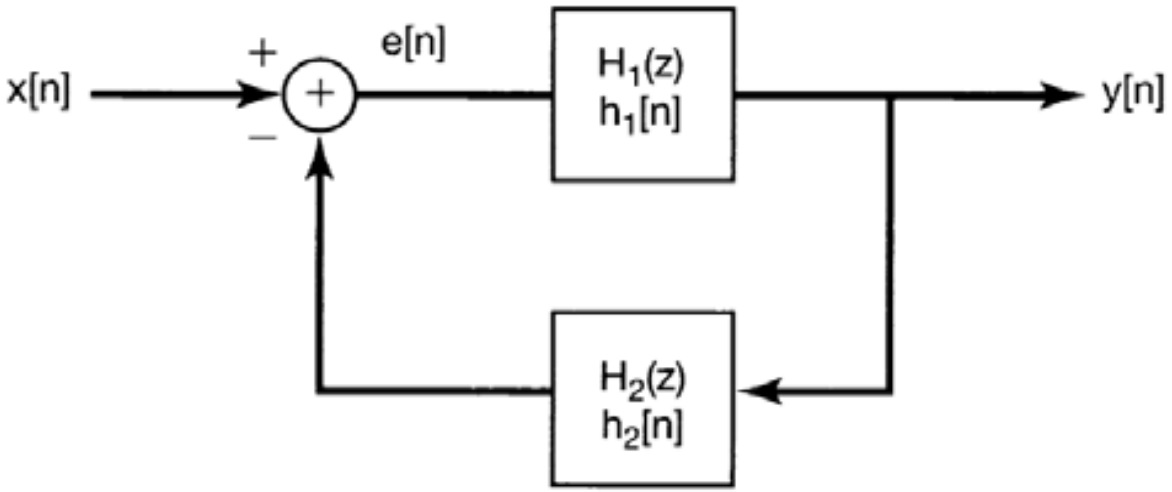
$$z_1=0.4 \quad p_1=0.5500+0.9526j \quad |p_1|=1.1>1$$

$$z_2=-0.4 \quad p_2=0.5500-0.9526j \quad |p_2|=1.1>1$$

2.9. Block Diagram

Feedback Block Diagram

$$\frac{Y(z)}{X(z)} = H(z) = \frac{H_1(z)}{1 + H_1(z)H_2(z)}$$



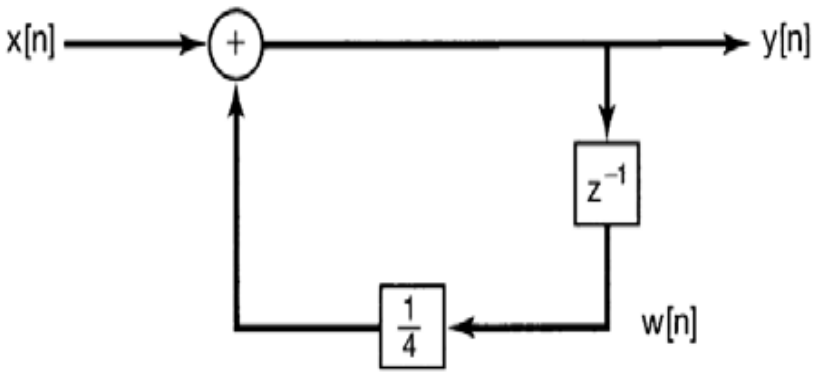
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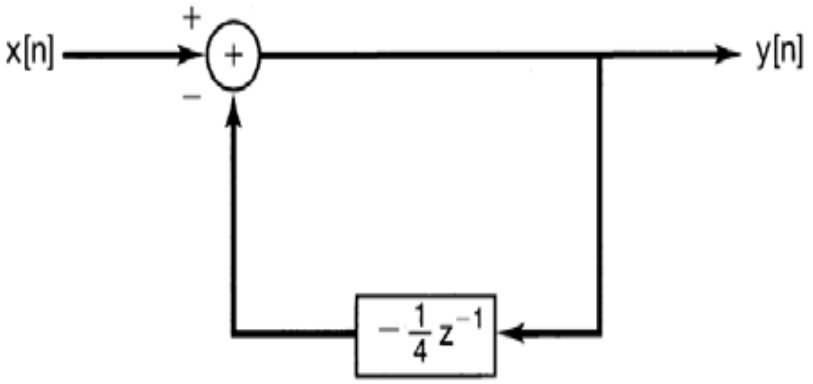
The Z-Transform

2.9. Block Diagram

$$H(z) = \frac{1}{1 - \frac{1}{4}z^{-1}}$$

$$y[n] - \frac{1}{4}y[n-1] = x[n]$$





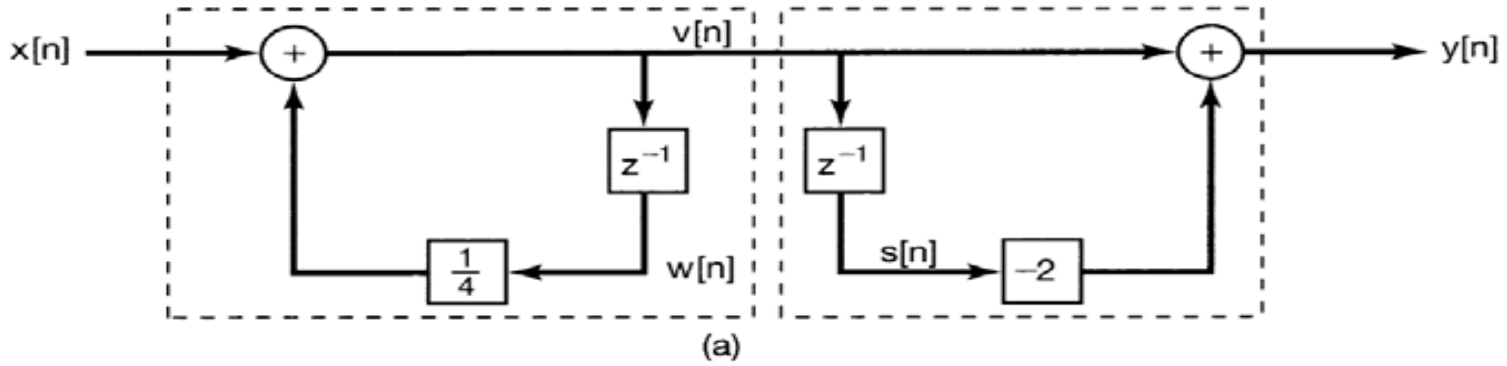
Digital Signal Processing

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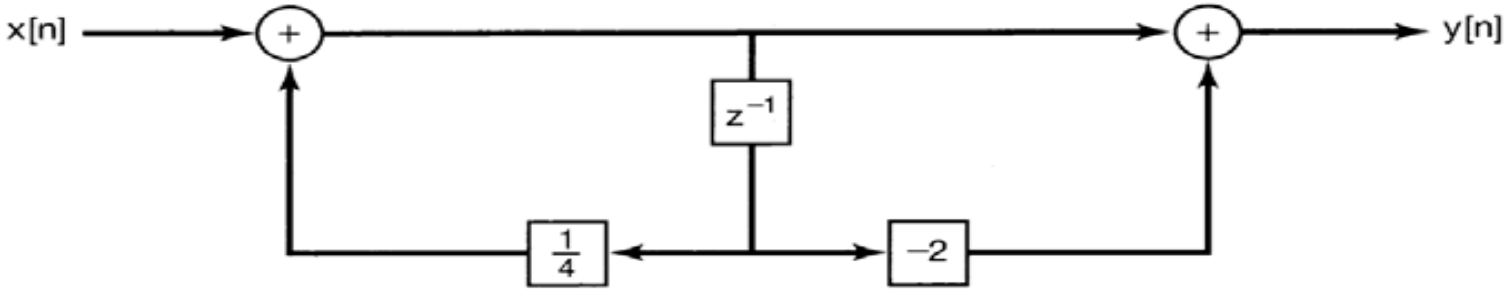
The Z-Transform

2.9. Block Diagram

$$H(z) = \frac{1 - 2z^{-1}}{1 - \frac{1}{4}z^{-1}} = \left(\frac{1}{1 - \frac{1}{4}z^{-1}}\right)(1 - 2z^{-1})$$



(a)



(b)

2

The Z-Transform

2.10. Comparing between Domains

Time – Domain: impulse response $h(k)$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k} \qquad H(z) = \sum_{k=-\infty}^{\infty} h(k)z^{-k}$$

***Z* – Domain:** transfer function $H(z)$

$$H\left(e^{j\omega}\right) = H\left(z\right)_{z=e^{jw}} \qquad h(n) = \frac{1}{2\pi j}\oint_C H(z)z^{n-1}dz$$

Frequency – Domain: frequency response $H\left(e^{j\omega}\right)$

$$H(z) = H\left(e^{j\omega}\right)_{e^{j\omega}=z} \qquad h(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(e^{j\omega})e^{j\omega k}d\omega$$